

1. $A = 3x^2 - 5x - 3$, $B = x^2 + 2x - 3$, $C = -3x^2 - 4x + 1$ であるとき, 次の計算をせよ。

(1) $-A + B$

答. _____

(2) $3B + 2C$

答. _____

(3) $A + B - C$

答. _____

(4) $-A - B + C$

答. _____

(5) $-3A - 2B - C$

答. _____

(6) $2A - B + 3C$

答. _____

(7) $-2A - B - C$

答. _____

(8) $A - 2B + C$

答. _____

2. 次の 1 次方程式を解け。

(1) $9x - 4 = 4x + 7$

答. _____

(2) $-8x + 1 = 7x + 1$

答. _____

(3) $8x - 7 - (-6x - 7) = 0$

答. _____

(4) $-9x - 2(9x + 4) = -2$

答. _____

(5) $9x + 5 + 4(3x + 9) = 0$

答. _____

(6) $-2x + 5 = \frac{9}{2}x + \frac{4}{3}$

答. _____

(7) $\frac{3}{2}x - \frac{3}{2} = 4x + 3$

答. _____

(8) $\frac{5}{2}x - \frac{5}{4} = \frac{7}{4}x + \frac{5}{3}$

答. _____

1. $A = 3x^2 - 5x - 3$, $B = x^2 + 2x - 3$, $C = -3x^2 - 4x + 1$ であるとき, 次の計算をせよ。

(1) $-A + B$

答. $-2x^2 + 7x$

(2) $3B + 2C$

答. $-3x^2 - 2x - 7$

(3) $A + B - C$

答. $7x^2 + x - 7$

(4) $-A - B + C$

答. $-7x^2 - x + 7$

(5) $-3A - 2B - C$

答. $-8x^2 + 15x + 14$

(6) $2A - B + 3C$

答. $-4x^2 - 24x$

(7) $-2A - B - C$

答. $-4x^2 + 12x + 8$

(8) $A - 2B + C$

答. $-2x^2 - 13x + 4$

2. 次の 1 次方程式を解け。

(1) $9x - 4 = 4x + 7$

$5x = 11$

答. $x = \frac{11}{5}$

(2) $-8x + 1 = 7x + 1$

$-15x = 0$

答. $x = 0$

(3) $8x - 7 - (-6x - 7) = 0$

$8x - 7 + 6x + 7 = 0$

$14x = 0$

答. $x = 0$

(4) $-9x - 2(9x + 4) = -2$

$-9x - 18x - 8 = -2$

$-27x = 6$

答. $x = -\frac{2}{9}$

(5) $9x + 5 + 4(3x + 9) = 0$

$9x + 5 + 12x + 36 = 0$

$21x = -41$

答. $x = -\frac{41}{21}$

(6) $-2x + 5 = \frac{9}{2}x + \frac{4}{3}$

両辺に 6 をかけて, $-12x + 30 = 27x + 8$

答. $x = \frac{22}{39}$

(7) $\frac{3}{2}x - \frac{3}{2} = 4x + 3$

両辺に 2 をかけて, $3x - 3 = 8x + 6$

答. $x = -\frac{9}{5}$

(8) $\frac{5}{2}x - \frac{5}{4} = \frac{7}{4}x + \frac{5}{3}$

両辺に 12 をかけて, $30x - 15 = 21x + 20$

答. $x = \frac{35}{9}$